# Numerical analysis of superconducting phases in the extended Hubbard model with non-local pairing

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**Abstract**

[To be continued...]

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**List of symbols and abbreviations**

| | |
|---|---|
| AF | Antiferromagnetic |
| BCS | Bardeen-Cooper-Schrieffer (theory) |
| $\beta$ | Inverse temperature |
| $\Delta^{(\gamma)}$ | $\gamma$-wave component of the gap function |
| DoF | Degree(s) of freedom |
| EHM | Extended Hubbard model |
| EV(s) | Expectation value(s) |
| FS | Fermi surface |
| HF | Hartree-Fock |
| HFP(s) | Hartree-Fock parameter(s) |
| HM | Hubbard model |
| $L_\ell$ | Number of lattice sites in direction $\ell$ |
| LRT | Linear Response Theory |
| MFT | Mean-Field Theory |
| NN(s) | Nearest neighbor(s) |
| RWC(s) | Relevant Wick's contraction(s) |
| RMP(s) | Renormalized model parameter(s) |
| SC | Superconductor |
| SDW | Spin-density wave |
| SSB | Spontaneous symmetry breaking |
| $T_c$ | Critical temperature |

# Chapter 1

# Superconducting instability

This chapter is devoted to studying the superconducting phase of the system. The only symmetry we assume to break is the $\mathrm{U}^{\mathrm{c}}(1)$ charge symmetry, thus allowing for superconducting fluctuations. As is described thoroughly in Sec. 1.3.1, the hopping amplitude is renormalized because of the non-local attraction. The symmetry structure of the pairing mechanism determines the contributing Cooper fluctuations: for $s$-wave and $d$-wave superconductivity, only the o.s. Cooper term contributes; for $p_\ell$-wave superconductivity, the s.s. term contributes as well. In the following sections, a derivation containing both Cooper terms is proposed.

## 1.1 The SC phase and its symmetries

The starting point is, as done for the other phases, the discussion of the symmetries of the SC phase. We said in introduction we are studying the

$$\mathrm{SU}(2) \times \mathrm{U}^{\mathrm{c}}(1) \times (\text{Point group}) \quad \overset{\text{SSB}}{\to} \quad \mathrm{SU}(2) \times (\text{Point group})$$

symmetry breaking. Therefore the system density we assume to be uniform across all sites, similarly to the normal phase discussed in Chap. **??**. [To be continued...]

Let us highlight now an important point regarding the Cooper class Wick's contractions performed on the general EHM of Eq. (**??**),

$$\hat{H} = \underbrace{-t \sum_{\langle ij \rangle} \sum_{\sigma} \hat{c}^\dagger_{i\sigma} \hat{c}_{j\sigma}}_{\hat{H}_t} \underbrace{+U \sum_{i} \hat{c}^\dagger_{i\uparrow} \hat{c}^\dagger_{i\downarrow} \hat{c}_{i\downarrow} \hat{c}_{i\uparrow}}_{\hat{H}_U} \underbrace{-V \sum_{\langle ij \rangle} \sum_{\sigma\sigma'} \hat{c}^\dagger_{i\sigma} \hat{c}^\dagger_{j\sigma'} \hat{c}_{j\sigma'} \hat{c}_{i\sigma}}_{\hat{H}_V}$$

As is discussed in Sec. **??**, when applying Wick's theorem the resulting terms break the natural symmetries of the model. The superconducting symmetry we study breaks just the $\mathrm{U}^{\mathrm{c}}(1)$ charge conservation. Now, when dealing with superconducting Cooper pairing we need to account also for the spatial structure of the Cooper pair itself. Consider the generic Cooper fluctuation

$$\langle \hat{c}^\dagger_{i\sigma} \hat{c}^\dagger_{j\sigma'} \rangle \quad \text{with } i, j \text{ NN}$$

From basic Quantum Mechanics we know the summation rules of the spin algebra $\mathfrak{su}(2)$,

$$\frac{1}{2} \otimes \frac{1}{2} = 0 \oplus 1$$

The two pairing channels are, at this level, the singlet channel associated to total spin 0 and the triplet channel associated to total spin 1. If we impose a specific spatial symmetry on the hamiltonian the ground state wavefunction will follow naturally, and the pairing channel will be the one providing a total anti-symmetry to the full wavefunction. This gives a selection rule over the relevant pairings: if we work with space symmetric structures – say, $s^*$-wave or $d$-wave – the pairing will take place in singlet channel, allowing us to eliminate the triplet pairing. Working with antisymmetric structures, inversely, we can eliminate the singlet channel. This concept is summarized in Tab. 1.1.

| Spatial structure | Pairing channel | RWCs in Cooper channel |
|---|---|---|
| Symmetric wave function | Singlet pairing | Just $\langle \hat{c}^\dagger_{i\sigma}\hat{c}^\dagger_{j\overline{\sigma}}\rangle$ |
| Anti-symmetric wave function | Triplet pairing | Both $\langle \hat{c}^\dagger_{i\sigma}\hat{c}^\dagger_{j\overline{\sigma}}\rangle$ and $\langle \hat{c}^\dagger_{i\sigma}\hat{c}^\dagger_{j\sigma}\rangle$ |

Table 1.1 Relation of the Cooper pairing channel with the wavefunction spatial symmetry (intended as the inversion $(x, y) \to (-x, -y)$).

### 1.1.1 Superconductivity in the HM

In Sec. **??**, we discuss in detail the (impossibility of a stable) $s$-wave SC phase in the conventional HM. This is an obvious result, since the two-body interaction thereby present $\hat{H}_U$ is local and repulsive. We nevertheless treat the model within the MFT framework, and here recall the main widely known results, and later use them as groundwork for the EHM extension. Similarly to Sec. **??**, let us define the operators

$$\hat{n}_{\mathbf{k}\sigma} \equiv \hat{c}^\dagger_{\mathbf{k}\sigma}\hat{c}_{\mathbf{k}\sigma} \tag{1.1}$$

$$\hat{\phi}_{\mathbf{k}} \equiv \hat{c}^\dagger_{\mathbf{k}\uparrow}\hat{c}^\dagger_{-\mathbf{k}\downarrow} \tag{1.2}$$

whose EVs are relevant for the SC phase. Let:

$$u_{\mathbf{q}\sigma} \equiv \langle \hat{n}_{\mathbf{q}\sigma}\rangle \tag{1.3}$$

$$w_{\mathbf{q}\sigma} \equiv \langle \hat{\phi}^\dagger_{\mathbf{q}}\rangle \tag{1.4}$$

The MFT hamiltonian is given explicitly for the HM in Eq. (**??**),

$$\hat{H}_t + \hat{H}_U \overset{\text{MFT}}{\simeq} \sum_{\mathbf{k}\in\text{BZ}}\sum_{\sigma}\epsilon_{\mathbf{k}\sigma}\left[\hat{c}^\dagger_{\mathbf{k}\sigma}\hat{c}_{\mathbf{k}\sigma} - \hat{c}^\dagger_{\mathbf{k}+\boldsymbol{\pi}\sigma}\hat{c}_{\mathbf{k}+\boldsymbol{\pi}\sigma}\right] - \sum_{\mathbf{q}\in\text{BZ}}\left[\Delta^{(s)}\hat{\phi}_{\mathbf{q}} + \Delta^{(s)*}\hat{\phi}^\dagger_{\mathbf{q}}\right] + nU\times\hat{N} - \left[E^{(\text{SC})}_{\text{H}/U} + E^{(\text{SC})}_{\text{C}/U}\right]$$

with $E^{(\text{SC})}_{\text{H}/U}$, $E^{(\text{SC})}_{\text{C}/U}$ the contraction energies arising from Hartree and Cooper channels, respectively in Eq. (**??**) and (**??**). In deriving this result we decomposed the interaction of the intermediate passage of Eq. (**??**) and assumed the Cooper pair to have null total momentum. Moreover, we recognized in the passage of Eq. (**??**) the presence of a $s$-wave harmonic transform of the function $w_{\mathbf{q}}$, and thus defined in Eq. (**??**),

$$w^{(s)} \equiv \frac{1}{L_xL_y}\sum_{\mathbf{q}\in\text{BZ}} w_{\mathbf{q}} \quad \text{and} \quad \Delta^{(s)} \equiv -Uw^{(s)}$$

Nambu-Bogoliubov representation of the MFT hamiltonian is given in Eq. (**??**) and equivalently in Eq. (**??**),

$$\hat{H}_t + \hat{H}_U - \mu\hat{N} \simeq \sum_{\mathbf{k}\in\text{BZ}}\sum_{\sigma}\hat{\Phi}^\dagger_{\mathbf{k}}h_{\mathbf{k}}\hat{\Phi}_{\mathbf{k}} - \left[E^{(\text{SC})}_{\text{A}/t} + E^{(\text{SC})}_{\text{H}/U} + E^{(\text{SC})}_{\text{C}/U}\right]$$

being

$$h_{\mathbf{k}} \equiv \begin{bmatrix}\xi_{\mathbf{k}} & -\Delta_{\mathbf{k}}\\ -\Delta^*_{\mathbf{k}} & -\xi_{\mathbf{k}}\end{bmatrix} \quad \text{and} \quad \xi_{\mathbf{k}} \equiv \epsilon_{\mathbf{k}} - \tilde{\mu}$$

and $\Delta_{\mathbf{k}} = \Delta^{(s)}$. Nambu spinors are, in this case,

$$\hat{\Phi}_{\mathbf{k}} \equiv \begin{bmatrix}\hat{c}_{\mathbf{k}\uparrow}\\ \hat{c}^\dagger_{-\mathbf{k}\downarrow}\end{bmatrix}$$

For the SC phase in the HM model, in general, we can admit for a complex gap function – although being the gap a constant it can always be reduced to a purely real gap. Diagonalization by the angles defined in Eq. (**??**) is performed via the composed rotation of Eq. (**??**),

$$W_{\mathbf{k}} = \begin{bmatrix}\cos\theta_{\mathbf{k}}e^{-i\zeta_{\mathbf{k}}} & -\sin\theta_{\mathbf{k}}e^{i\zeta_{\mathbf{k}}}\\ \sin\theta_{\mathbf{k}}e^{-i\zeta_{\mathbf{k}}} & \cos\theta_{\mathbf{k}}e^{i\zeta_{\mathbf{k}}}\end{bmatrix}$$

| EV | Meaning | Equation |
|---|---|---|
| $\langle \hat{s}^x_{\mathbf{k}} \rangle$ | $\left\langle \hat{\phi}_{\mathbf{k}} + \hat{\phi}^\dagger_{\mathbf{k}} \right\rangle$ | (**??**) |
| $i \langle \hat{s}^y_{\mathbf{k}} \rangle$ | $\left\langle \hat{\phi}_{\mathbf{k}\sigma} - \hat{\phi}^\dagger_{\mathbf{k}} \right\rangle$ | (**??**) |
| $\langle \hat{s}^z_{\mathbf{k}} \rangle$ | $\langle \hat{n}_{\mathbf{k}\uparrow} + \hat{n}_{\mathbf{k}\downarrow} \rangle - 1$ | (**??**) |

Table 1.2 List of the various expectation values (EVs) of each pseudospin component, their meaning in terms of fermionic operators and the reference equations from the pure HM. The $z$ component refers also to Eq. (**??**).

The Bogoliubov bands are $\pm E_{\mathbf{k}} = \sqrt{\xi^2_{\mathbf{k}} + |\Delta_{\mathbf{k}}|^2}$. The only crucial difference with the AF Bogoliubov bands it that here, apart from the absence of the spin index – a degeneracy that is solved now by enlarging the computation to the entire BZ –, also the bare bands $\epsilon_{\mathbf{k}}$ are substituted by their shifted version $\xi_{\mathbf{k}}$. This difference is essential: it is this feature that makes the superconducting phase gapped in an insulating sense, which means, opening the gap at the position of $\tilde{\mu}$. Instead for the AF phase the gap was always opened at "midband", the MBZ contour where $\epsilon_{\mathbf{k}} = 0$. As a final remark, as is proven trivially at the end of Sec. **??**, $s$-wave SC is impossible at MFT level on the pure HM. This was obvious, due to the necessity of an even small attractive interaction to form Cooper pairs. To conclude, in Tab. 1.2 we report the operator-pseudospin map for each direction.

### 1.1.2 Extension to the EHM

Add now the non-local interaction $\hat{H}_V$: translational invariance is preserved while charge conservation is broken, thus all three channels (Hartree, Fock and Cooper) contribute via RWCs to the MFT hamiltonian. As for the AF phase, we aim to derive a MFT solution for the SC phase which is an extension of the HM solution of Sec. **??**, with an effective redefinition of the various physical quantities and mathematical objects,

$$\xi_{\mathbf{k}} \to \tilde{\xi}_{\mathbf{k}} \qquad E_{\mathbf{k}} \to \tilde{E}_{\mathbf{k}} \qquad \Delta_{\mathbf{k}} \to \tilde{\Delta}_{\mathbf{k}} \qquad \mu \to \tilde{\mu}$$

To be completely correct, the chemical potential was already renormalized in the HM derivation in Sec. **??**. It here acquires yet another renormalization. The band energies renormalization is simply

$$\tilde{E}_{\mathbf{k}} \equiv \sqrt{\tilde{\xi}^2_{\mathbf{k}} + |\tilde{\Delta}_{\mathbf{k}}|^2}$$

and as for the AF phase:

$$\langle \hat{\Psi}^\dagger_{\mathbf{k}\sigma} \tau^x \hat{\Psi}_{\mathbf{k}\sigma} \rangle = -\sin(2\tilde{\theta}_{\mathbf{k}\sigma}) \sin(2\tilde{\zeta}_{\mathbf{k}\sigma}) \langle \hat{\Gamma}^\dagger_{\mathbf{k}\sigma} \tau^z \hat{\Gamma}_{\mathbf{k}\sigma} \rangle \tag{1.5}$$

$$\langle \hat{\Psi}^\dagger_{\mathbf{k}\sigma} \tau^y \hat{\Psi}_{\mathbf{k}\sigma} \rangle = -\sin(2\tilde{\theta}_{\mathbf{k}\sigma}) \cos(2\tilde{\zeta}_{\mathbf{k}\sigma}) \langle \hat{\Gamma}^\dagger_{\mathbf{k}\sigma} \tau^z \hat{\Gamma}_{\mathbf{k}\sigma} \rangle \tag{1.6}$$

$$\langle \hat{\Psi}^\dagger_{\mathbf{k}\sigma} \tau^z \hat{\Psi}_{\mathbf{k}\sigma} \rangle = \cos(2\tilde{\theta}_{\mathbf{k}\sigma}) \langle \hat{\Gamma}^\dagger_{\mathbf{k}\sigma} \tau^z \hat{\Gamma}_{\mathbf{k}\sigma} \rangle \tag{1.7}$$

These equations are identical to the ones for the pure HM, Eqns. (**??**), (**??**) and (**??**). In the tilted frame, we have the analogous of Eq. (**??**) with the important exception that the chemical potential in the Fermi-Dirac factors is zero,

$$\langle \hat{\Gamma}^\dagger_{\mathbf{k}\sigma} \tau^z \hat{\Gamma}_{\mathbf{k}\sigma} \rangle = \frac{1}{2} \left[ f\left(\tilde{E}_{\mathbf{k}\sigma}; \beta, 0\right) - f\left(-\tilde{E}_{\mathbf{k}\sigma}; \beta, 0\right) \right] \tag{1.8}$$

This is because all the chemical potential operatorial contributions were absorbed in the bare bands $\tilde{\xi}_{\mathbf{k}}$, thus the pseudospin state is thermally described by an effective null chemical potential.

The (renormalized) thermal factors as in Eq. (**??**) are

$$\tilde{\mathrm{Th}}^{(\pm)}_{\mathbf{k}\sigma}[\beta, \tilde{\mu}] \equiv f\left(\tilde{E}_{\mathbf{k}\sigma}; \beta, 0\right) \pm f\left(-\tilde{E}_{\mathbf{k}\sigma}; \beta, 0\right) \tag{1.9}$$

while for the angles it holds

$$\sin(2\tilde{\xi}_{\mathbf{k}\sigma}) = \frac{\mathrm{Re}\{\tilde{\Delta}_{\mathbf{k}\sigma}\}}{|\tilde{\Delta}_{\mathbf{k}\sigma}|} \quad \cos(2\tilde{\xi}_{\mathbf{k}\sigma}) = \frac{\mathrm{Im}\{\tilde{\Delta}_{\mathbf{k}\sigma}\}}{|\tilde{\Delta}_{\mathbf{k}\sigma}|} \quad \sin(2\tilde{\theta}_{\mathbf{k}\sigma}) = \frac{|\tilde{\Delta}_{\mathbf{k}\sigma}|}{\tilde{E}_{\mathbf{k}\sigma}} \quad \cos(2\tilde{\theta}_{\mathbf{k}\sigma}) = \frac{\tilde{\xi}_{\mathbf{k}\sigma}}{\tilde{E}_{\mathbf{k}\sigma}} \tag{1.10}$$

From Tabs. **??** and **??**, we know all channel contribute. It is already known the effect of the Fock contractions, responsible as for the normal phase (Chap. **??**) of the bare bands renormalization. Cooper terms are to be handled more carefully, since can contribute to superconductivity both in the s.s. and in the o.s. spin sectors.

## 1.2 Non-local Hartree effects: chemical potential renormalization

As always, let us start with a very brief section about the Hartree effects on the MFT decomposition of the hamiltonian. We will not delve in the detailed calculations of the Hartree channel MFT effects, already described in previous chapter as well as in Sec. **??**. The only effect here is the chemical potential renormalization.

### 1.2.1 Hartree channel in the SC phase: analytical derivation

The Hartree fluctuations are density-density couplings in the MFT hamiltonian. Knowing the deal with a uniform-density phase, we can re-use as it is the derivation of Sec. **??**. Without repeating the calculations, the $\hat{H}_V$ is reduced in the Hartree channel to

$$\hat{H}_V \simeq -2nzV \times \hat{N} - E^{(\mathrm{SC})}_{\mathrm{H}/V} + (\text{all the rest})$$

This, together with the $\hat{H}_U$ Hartree channel decomposition

$$\hat{H}_U \simeq nU \times \hat{N} - E^{(\mathrm{SC})}_{\mathrm{H}/U}$$

gives the chemical potential full renormalization, identical to Eq. (**??**),

$$\tilde{\mu} \equiv \mu + n(2zV - U) \tag{1.11}$$

### 1.2.2 Hartree contraction energy

The contraction energy for the non-local sector was derived in Eq. (**??**),

$$E^{(\mathrm{SC})}_{\mathrm{H}/V} = -L_x L_y \times 2zn^2 V$$

and together with the local counterpart of Eq. (**??**)

$$E^{(\mathrm{SC})}_{\mathrm{H}/U} = L_x L_y \times n^2 U$$

Now we move to the more interesing Fock sector.

## 1.3 Fock effects: bare bands renormalization

[To be continued...]

### 1.3.1 Fock channel in the SC phase: analytical derivation

When it comes to the Fock RWCs, being the SC phase density-uniform we can re-use almost the entirety of the derivation of Sec. **??**. Setting as in Eq. (**??**)

$$u_{\mathbf{q}\sigma} \equiv \langle \hat{c}^\dagger_{\mathbf{q}\sigma} \hat{c}_{\mathbf{q}\sigma} \rangle \tag{1.12}$$

Eq. (**??**) stands perfectly valid, translated to the SC phase

$$V\sum_{\langle ij\rangle}\sum_{\sigma}\left[\langle\hat{c}^{\dagger}_{i\sigma}\hat{c}_{j\sigma}\rangle\hat{c}^{\dagger}_{j\sigma}\hat{c}_{i\sigma}+\hat{c}^{\dagger}_{i\sigma}\hat{c}_{j\sigma}\langle\hat{c}^{\dagger}_{j\sigma}\hat{c}_{i\sigma}\rangle-\langle\hat{c}^{\dagger}_{i\sigma}\hat{c}_{j\sigma}\rangle\langle\hat{c}^{\dagger}_{j\sigma}\hat{c}_{i\sigma}\rangle\right]$$
$$\overset{\mathcal{F}}{=}\sum_{\gamma,\sigma}\left[\frac{1}{L_xL_y}\sum_{\mathbf{q}\in\mathrm{BZ}}\varphi^{(\gamma)}_{\mathbf{q}}u_{\mathbf{q}\sigma}\right]\times V\sum_{\mathbf{q}'\in\mathrm{BZ}}\varphi^{(\gamma)*}_{\mathbf{q}'}\hat{c}^{\dagger}_{\mathbf{q}'\sigma}\hat{c}_{\mathbf{q}'\sigma}-E^{(\mathrm{SC})}_{\mathrm{F}/V} \tag{1.13}$$

being

$$E^{(\mathrm{SC})}_{\mathrm{F}/V}\equiv\frac{V}{L_xL_y}\sum_{\mathbf{q},\mathbf{q}'}\sum_{\sigma}\left[\cos\left(\delta q_x\right)+\cos\left(\delta q_y\right)\right]\langle\hat{c}^{\dagger}_{\mathbf{q}\sigma}\hat{c}_{\mathbf{q}\sigma}\rangle\langle\hat{c}^{\dagger}_{\mathbf{q}'\sigma}\hat{c}_{\mathbf{q}'\sigma}\rangle \tag{1.14}$$

To get to this point, we employed the assumption

$$\langle\hat{c}^{\dagger}_{\mathbf{k}\sigma}\hat{c}_{\mathbf{k}'\sigma'}\rangle=\delta_{\mathbf{k}=\mathbf{k}'}\delta_{\sigma=\sigma'}\langle\hat{c}^{\dagger}_{\mathbf{k}\sigma}\hat{c}_{\mathbf{k}\sigma}\rangle$$

which essentially states translational invariance and spin degeneracy. This assumption is obviously well-verified by the BCS ground state of Eq. [Missing]. Recognising the symmetry decomposition

$$u^{(\gamma)}_{\sigma}=\frac{1}{L_xL_y}\sum_{\mathbf{q}\in\mathrm{BZ}}\varphi^{(\gamma)}_{\mathbf{q}}\langle\hat{c}^{\dagger}_{\mathbf{q}\sigma}\hat{c}_{\mathbf{q}\sigma}\rangle$$

due to inversion symmetry the conclusion is identical to Eqns. (**??**) and (**??**) (considering spin degeneracy),

$$\tilde{\epsilon}_{\mathbf{k}}\equiv\epsilon_{\mathbf{k}}+u^{(s^*)}V(\cos q_x+\cos q_y)+u^{(d)}V(\cos q_x-\cos q_y)$$

In order to estimate the $\gamma$-wave components renormalizing the bare bands, we need to compute separately in spin sectors the EVs of $u_{\mathbf{k}\sigma}$

$$\begin{aligned}
u_{\mathbf{k}\uparrow}&=\langle\hat{c}^{\dagger}_{\mathbf{k}\uparrow}\hat{c}_{\mathbf{k}\uparrow}\rangle \qquad (1.15)\\
&=\left\langle\hat{\Phi}^{\dagger}_{\mathbf{k}}\frac{\mathbb{I}+\tau^z}{2}\hat{\Phi}_{\mathbf{k}}\right\rangle\\
&=\frac{1}{2}\left[\left\langle\hat{\Gamma}^{\dagger}_{\mathbf{k}}\hat{\Gamma}_{\mathbf{k}}\right\rangle+\langle\hat{s}^z_{\mathbf{k}}\rangle\right]\\
&=\frac{1}{2}\left[\tilde{\mathrm{Th}}^{(+)}_{\mathbf{k}}[\beta]+\cos 2\tilde{\theta}_{\mathbf{k}}\tilde{\mathrm{Th}}^{(-)}_{\mathbf{k}}[\beta]\right]\\
&=\frac{1+\cos 2\tilde{\theta}_{\mathbf{k}}}{2}f\left(\tilde{E}_{\mathbf{k}};\beta,0\right)+\frac{1-\cos 2\tilde{\theta}_{\mathbf{k}}}{2}f\left(-\tilde{E}_{\mathbf{k}};\beta,0\right)\\
&=\cos^2\tilde{\theta}_{\mathbf{k}}f\left(\tilde{E}_{\mathbf{k}};\beta,0\right)+\sin^2\tilde{\theta}_{\mathbf{k}}f\left(-\tilde{E}_{\mathbf{k}};\beta,0\right)\\
&=\sin^2\tilde{\theta}_{\mathbf{k}}\tanh\left(\frac{\beta\tilde{E}_{\mathbf{k}}}{2}\right)+f\left(\tilde{E}_{\mathbf{k}};\beta,0\right) \qquad (1.16)
\end{aligned}$$

having we used the simple relation

$$\begin{aligned}
\frac{1}{e^{-x}+1}-\frac{1}{e^{x}+1}&=\frac{e^x-1}{e^x+1}\\
&=\tanh\left(\frac{x}{2}\right)
\end{aligned}$$

For the spin $\downarrow$ EV,

$$\begin{aligned}
u_{-\mathbf{k}\downarrow} &= \langle \hat{c}^\dagger_{-\mathbf{k}\downarrow} \hat{c}_{-\mathbf{k}\downarrow} \rangle \\
&= 1 - \langle \hat{c}_{-\mathbf{k}\downarrow} \hat{c}^\dagger_{-\mathbf{k}\downarrow} \rangle \\
&= 1 - \left\langle \hat{\Phi}^\dagger_{\mathbf{k}} \frac{\mathbb{I} - \tau^z}{2} \hat{\Phi}_{\mathbf{k}} \right\rangle \\
&= 1 - \frac{1}{2} \left[ \left\langle \hat{\Gamma}^\dagger_{\mathbf{k}} \hat{\Gamma}_{\mathbf{k}} \right\rangle - \langle \hat{s}^z_{\mathbf{k}} \rangle \right] \\
&= 1 - \frac{1}{2} \left[ \tilde{\mathrm{Th}}^{(+)}_{\mathbf{k}}[\beta] - \cos 2\tilde{\theta}_{\mathbf{k}} \tilde{\mathrm{Th}}^{(-)}_{\mathbf{k}}[\beta] \right] \\
&= 1 - \frac{1 - \cos 2\tilde{\theta}_{\mathbf{k}}}{2} f\left(\tilde{E}_{\mathbf{k}}; \beta, 0\right) - \frac{1 + \cos 2\tilde{\theta}_{\mathbf{k}}}{2} f\left(-\tilde{E}_{\mathbf{k}}; \beta, 0\right) \\
&= 1 - \sin^2 \tilde{\theta}_{\mathbf{k}} f\left(\tilde{E}_{\mathbf{k}}; \beta, 0\right) - \cos^2 \tilde{\theta}_{\mathbf{k}} f\left(-\tilde{E}_{\mathbf{k}}; \beta, 0\right) \\
&= 1 - \sin^2 \tilde{\theta}_{\mathbf{k}} \tanh\left(\frac{\beta \tilde{E}_{\mathbf{k}}}{2}\right) - f\left(-\tilde{E}_{\mathbf{k}}; \beta, 0\right) \\
&= -\sin^2 \tilde{\theta}_{\mathbf{k}} \tanh\left(\frac{\beta \tilde{E}_{\mathbf{k}}}{2}\right) + f\left(\tilde{E}_{\mathbf{k}}; \beta, 0\right)
\end{aligned} \tag{1.17}$$

These results will be used to extract self-consistency equations for each HFP. Note that

$$\lim_{|\Delta_{\mathbf{k}}| \to 0} u_{\mathbf{k}\sigma} = f(\tilde{\xi}_{\mathbf{k}}; \beta, 0)$$

Correctly, the null-gap limit reduces to a perfectly degenerate Fermi gas.

### 1.3.2 Fock contraction energy

In terms of the energy shift due to Wick's contraction, Eq. (**??**) holds identically

$$E^{(\mathrm{SC})}_{\mathrm{F}/V} = L_x L_y \times \frac{V}{2} \sum_\sigma \left[ \left| u^{(s^*)} \right|^2 + \left| u^{(d)} \right|^2 \right]$$

with the only obvious difference that the $u^{(\gamma)}$ are computed on the SC ground state by the means of Eqns. (1.16), (1.17).

## 1.4 Cooper channel in the o.s. and s.s. sectors

Let us get back to the Fourier transform of Eq. (**??**)

$$\hat{H}_V \overset{\mathcal{F}}{=} -\frac{V}{L_x L_y} \sum_{\sigma\sigma'} \sum_{\mathbf{K},\mathbf{k},\mathbf{k}'} \left[\cos(\delta k_x) + \cos(\delta k_y)\right] \hat{c}^\dagger_{\mathbf{K}+\mathbf{k}\sigma} \hat{c}^\dagger_{\mathbf{K}-\mathbf{k}\sigma'} \hat{c}_{\mathbf{K}-\mathbf{k}'\sigma'} \hat{c}_{\mathbf{K}+\mathbf{k}'\sigma}$$

which decomposes in s.s. and o.s. channels as:

$$\begin{aligned}
\hat{H}_V \overset{\mathcal{F}}{=} &\overbrace{-\frac{V}{L_x L_y} \sum_{\sigma} \sum_{\mathbf{K},\mathbf{k},\mathbf{k}'} \left[\cos(\delta k_x) + \cos(\delta k_y)\right] \hat{c}^\dagger_{\mathbf{K}+\mathbf{k}\sigma} \hat{c}^\dagger_{\mathbf{K}-\mathbf{k}\sigma} \hat{c}_{\mathbf{K}-\mathbf{k}'\sigma} \hat{c}_{\mathbf{K}+\mathbf{k}'\sigma}}^{\text{s.s.}} \\
&\underbrace{-\frac{V}{L_x L_y} \sum_{\sigma} \sum_{\mathbf{K},\mathbf{k},\mathbf{k}'} \left[\cos(\delta k_x) + \cos(\delta k_y)\right] \hat{c}^\dagger_{\mathbf{K}+\mathbf{k}\sigma} \hat{c}^\dagger_{\mathbf{K}-\mathbf{k}\overline{\sigma}} \hat{c}_{\mathbf{K}-\mathbf{k}'\overline{\sigma}} \hat{c}_{\mathbf{K}+\mathbf{k}'\sigma}}_{\text{o.s.}}
\end{aligned}$$

Following Tab. 1.1, in the singlet pairing channel only the o.s. sector produces RWCs. The latter is made up of two identical terms: this is because it holds

$$\hat{c}^\dagger_{\mathbf{K}+\mathbf{k}\sigma} \hat{c}^\dagger_{\mathbf{K}-\mathbf{k}\overline{\sigma}} \hat{c}_{\mathbf{K}-\mathbf{k}'\overline{\sigma}} \hat{c}_{\mathbf{K}+\mathbf{k}'\sigma} = \hat{c}^\dagger_{\mathbf{K}-\mathbf{k}\overline{\sigma}} \hat{c}^\dagger_{\mathbf{K}+\mathbf{k}\sigma} \hat{c}_{\mathbf{K}+\mathbf{k}'\sigma} \hat{c}_{\mathbf{K}-\mathbf{k}'\overline{\sigma}}$$

due to fermionic anti-commutation rules, and since the composite transform

$$\mathbf{k} \to -\mathbf{k} \qquad \mathbf{k}' \to -\mathbf{k}'$$

leaves the form factor $\cos(\delta k_x) + \cos(\delta k_y)$ invariant, we reduce the spin sum to two times the $\sigma = \uparrow$ part,

$$\hat{H}_V^{(\text{o.s.})} = -\frac{2V}{L_x L_y} \sum_{\mathbf{K},\mathbf{k},\mathbf{k}'} \left[\cos(\delta k_x) + \cos(\delta k_y)\right] \hat{c}^\dagger_{\mathbf{K}+\mathbf{k}\uparrow} \hat{c}^\dagger_{\mathbf{K}-\mathbf{k}\downarrow} \hat{c}_{\mathbf{K}-\mathbf{k}'\downarrow} \hat{c}_{\mathbf{K}+\mathbf{k}'\uparrow} \tag{1.18}$$

[To be continued...]

## 1.5 Cooper effects in the singlet channel: gap renormalization

[To be continued...]

### 1.5.1 Cooper-singlet channel in the SC phase: analytical derivation

This section deals with Cooper fluctuations induced by the o.s. part of the non-local hamiltonian –referring to the notation of Eq. (**??**)– somewhat the simplest form of Cooper pairing. This sector contributes both to singlet pairing and triplet pairing. Considering Cooper fluctuations in the singlet channel, we need to break $\mathrm{U}^{\mathrm{c}}(1)$ symmetry imposing space inversion symmetry, while for the triplet anti-symmetry is required. Set as in Eq. (1.4) the EV

$$w_{\mathbf{q}} = \langle \hat{\phi}^\dagger_{\mathbf{q}} \rangle$$

Let us now break down the MFT discussion for the local and non-local interactions.

**Local repulsion.** The detailed treatment of $\hat{H}_U$ is discussed in Sec. **??**, in the context of conventional HM. We here just sketch the most important results. By contracting in the Cooper channel the Fourier transform of $\hat{H}_U$ of Eq. (**??**),

$$\hat{H}_U \overset{\mathcal{F}}{=} \frac{U}{L_x L_y} \sum_{\mathbf{K},\mathbf{k},\mathbf{k}'} \hat{c}^\dagger_{\mathbf{K}+\mathbf{k}\uparrow} \hat{c}^\dagger_{\mathbf{K}-\mathbf{k}\downarrow} \hat{c}_{\mathbf{K}-\mathbf{k}'\downarrow} \hat{c}_{\mathbf{K}+\mathbf{k}'\uparrow}$$

in the hypothesis that preserving space translations invariance the net momentum of the Cooper pair is null, as in Eq. (**??**) the local part of the hamiltonian is reduced to

$$\hat{H}_U \simeq (\text{Hartree channel}) + \frac{U}{L_x L_y} \sum_{\mathbf{q},\mathbf{q}'} \left[ w_{\mathbf{q}} \hat{\phi}_{\mathbf{q}'} + w^*_{\mathbf{q}'} \hat{\phi}^\dagger_{\mathbf{q}} \right] - E^{(\text{SC})}_{\text{C}/U} \tag{1.19}$$

being $E^{(\text{SC})}_{\text{C}/U}$ the Cooper contraction given in Eq. (**??**). The Hartree channel is discussed in Sec. **??**. Let now:

$$w^{(s)} \equiv \frac{1}{L_x L_y} \sum_{\mathbf{q} \in \text{BZ}} w_{\mathbf{q}}$$

which reduces Eq. (1.19) to

$$\hat{H}_U \simeq (\text{Hartree channel}) + U \sum_{\mathbf{q} \in \text{BZ}} \left[ w^{(s)} \hat{\phi}_{\mathbf{q}} + w^{(s)*} \hat{\phi}^\dagger_{\mathbf{q}} \right] - E^{(\text{SC})}_{\text{C}/U} \tag{1.20}$$

As in Eq. (**??**), this will produce the $s$-wave part of the gap function.

**Non-local attraction.** Consider Eq. (1.18), and perform Wick contractions, giving

$$\begin{aligned} \hat{H}_V^{(\text{o.s.})} \simeq -\frac{2V}{L_x L_y} \sum_{\mathbf{K},\mathbf{k},\mathbf{k}'} \left[\cos(\delta k_x) + \cos(\delta k_y)\right] \times \Big[ &\langle \hat{c}^\dagger_{\mathbf{K}+\mathbf{k}\uparrow} \hat{c}^\dagger_{\mathbf{K}-\mathbf{k}\downarrow} \rangle \hat{c}_{\mathbf{K}-\mathbf{k}'\downarrow} \hat{c}_{\mathbf{K}+\mathbf{k}'\uparrow} \\ &+ \hat{c}^\dagger_{\mathbf{K}+\mathbf{k}\uparrow} \hat{c}^\dagger_{\mathbf{K}-\mathbf{k}\downarrow} \langle \hat{c}_{\mathbf{K}-\mathbf{k}'\downarrow} \hat{c}_{\mathbf{K}+\mathbf{k}'\uparrow} \rangle - \langle \hat{c}^\dagger_{\mathbf{K}+\mathbf{k}\uparrow} \hat{c}^\dagger_{\mathbf{K}-\mathbf{k}\downarrow} \rangle \langle \hat{c}_{\mathbf{K}-\mathbf{k}'\downarrow} \hat{c}_{\mathbf{K}+\mathbf{k}'\uparrow} \rangle \Big] + (\text{all the rest}) \end{aligned}$$

Identical considerations as above hold, and just the $\mathbf{K} = \mathbf{0}$ term contributes. We have finally

$$\hat{H}_V^{(\text{o.s.})} \simeq -\frac{2V}{L_x L_y} \sum_{\mathbf{q},\mathbf{q}'} \left[\cos(\delta q_x) + \cos(\delta q_y)\right] \left[w_{\mathbf{q}} \hat{\phi}_{\mathbf{q}'} + w_{\mathbf{q}}^* \hat{\phi}_{\mathbf{q}'}^\dagger\right] - E_{\text{C}/V/\text{o.s.}}^{(\text{SC})} + (\text{all the rest}) \tag{1.21}$$

where the contraction energy is

$$E_{\text{C}/V/\text{o.s.}}^{(\text{SC})} \equiv -\frac{2V}{L_x L_y} \sum_{\mathbf{q},\mathbf{q}'} \left[\cos(\delta q_x) + \cos(\delta q_y)\right] w_{\mathbf{q}} w_{\mathbf{q}'}^* \tag{1.22}$$

Making use of the decomposition of Eq. (**??**), Eq. (1.21) becomes

$$\begin{aligned}
\hat{H}_V^{(\text{o.s.})} &\simeq -V \sum_\gamma \left[\frac{1}{L_x L_y} \sum_{\mathbf{q}} \varphi_{\mathbf{q}}^{(\gamma)} w_{\mathbf{q}}\right] \times \sum_{\mathbf{q}'} \varphi_{\mathbf{q}'}^{(\gamma)*} \hat{\phi}_{\mathbf{q}'} + \text{h.c.} - E_{\text{C}/V/\text{o.s.}}^{(\text{SC})} + (\text{all the rest}) \\
&= -V \sum_\gamma \sum_{\mathbf{q}\in\text{BZ}} w^{(\gamma)} \varphi_{\mathbf{q}}^{(\gamma)*} \hat{\phi}_{\mathbf{q}} + \text{h.c.} - E_{\text{C}/V/\text{o.s.}}^{(\text{SC})} + (\text{all the rest})
\end{aligned}$$

extending the symmetry sum to $\gamma \in \{s^*, p_x, p_y, d_{x^2-y^2}\}$. We defined

$$w^{(\gamma)} \equiv \frac{1}{L_x L_y} \sum_{\mathbf{q}} \varphi_{\mathbf{q}}^{(\gamma)} w_{\mathbf{q}}$$

### 1.5.2 Cooper contraction energy

[To be continued...]

## 1.6 Full MFT hamiltonian and anisotropic pairing

CONTINUE FROM HERE

Define the full gap function as the sum of both contributions,

$$\Delta_{\mathbf{k}} \equiv \mathcal{V}_{\mathbf{k}} - \mathcal{U}_{\mathbf{k}} \tag{1.23}$$

The full self-consistency equation is given by the simple combination of Eqns. (**??**) and (**??**),

$$\Delta_{\mathbf{k}} \equiv \sum_{\mathbf{k}'} \left[V^{(s)} + V_{\mathbf{k}\mathbf{k}'}\right] \langle \hat{\phi}_{\mathbf{k}'}^\dagger \rangle \quad \text{with} \quad V^{(s)} = -\frac{U}{2L_x L_y} \tag{1.24}$$

The gap function decomposes in symmetry channels as well,

$$\Delta_{\mathbf{k}} = \sum_\gamma \Delta^{(\gamma)} \varphi_{\mathbf{k}}^{(\gamma)}$$

If SC arises in a specific symmetry channel, $\Delta_{\mathbf{k}}$ will show the same symmetry. It follows, due to orthonormality and using Eq. (1.24),

$$\begin{aligned}
\Delta^{(\gamma)} &= \frac{1}{L_x L_y} \sum_{\mathbf{k}} \varphi_{\mathbf{k}}^{(\gamma)*} \Delta_{\mathbf{k}} \\
&= \frac{1}{L_x L_y} \sum_{\mathbf{k}} \varphi_{\mathbf{k}}^{(\gamma)*} \sum_{\mathbf{k}'} \left[V^{(s)} + V_{\mathbf{k}\mathbf{k}'}\right] \langle \hat{\phi}_{\mathbf{k}'}^\dagger \rangle \\
&= \frac{1}{L_x L_y} \sum_{\mathbf{k}} \varphi_{\mathbf{k}}^{(\gamma)*} \sum_{\mathbf{k}'\gamma'} V^{(\gamma')} \varphi_{\mathbf{k}}^{(\gamma')} \varphi_{\mathbf{k}'}^{(\gamma')*} \langle \hat{\phi}_{\mathbf{k}'}^\dagger \rangle \\
&= V^{(\gamma)} \sum_{\mathbf{k}} \varphi_{\mathbf{k}}^{(\gamma)*} \langle \hat{\phi}_{\mathbf{k}}^\dagger \rangle
\end{aligned} \tag{1.25}$$

This result provides a set of self-consistency equations for each symmetry channel, listed in Tab. 1.3. Notice that to reconstruct self-consistently the full $s$-wave phase transition, the actual gap function is given by

$$\Delta^{(s)} + \Delta^{(s^*)}(c_x + c_y)$$

The $s$-wave transition is the only one equipped of both the local and the non-local parts.

[To be continued...]

| Structure | Self-consistency equation | Graph |
|---|---|---|
| $s$-wave | $\Delta^{(s)} = -\frac{U}{2L_xL_y}\sum_{\mathbf{k}}\langle\hat{\phi}^\dagger_{\mathbf{k}}\rangle$ | Fig. ?? |
| Extended $s$-wave | $\Delta^{(s^*)} = \frac{V}{L_xL_y}\sum_{\mathbf{k}}(c_x+c_y)\langle\hat{\phi}^\dagger_{\mathbf{k}}\rangle$ | Fig. ?? |
| $p_x$-wave | $\Delta^{(p_x)} = -i\sqrt{2}\frac{V}{L_xL_y}\sum_{\mathbf{k}}s_x\langle\hat{\phi}^\dagger_{\mathbf{k}}\rangle$ | Fig. ?? |
| $p_y$-wave | $\Delta^{(p_y)} = -i\sqrt{2}\frac{V}{L_xL_y}\sum_{\mathbf{k}}s_y\langle\hat{\phi}^\dagger_{\mathbf{k}}\rangle$ | Fig. ?? |
| $d_{x^2-y^2}$-wave | $\Delta^{(d)} = \frac{V}{L_xL_y}\sum_{\mathbf{k}}(c_x-c_y)\langle\hat{\phi}^\dagger_{\mathbf{k}}\rangle$ | Fig. ?? |

Table 1.3 Symmetry resolved self-consistency equations for the MFT parameters $\Delta^{(\gamma)}$, based on Eq. (1.24) and (1.25). By computing $\langle\hat{\phi}^\dagger_{\mathbf{k}}\rangle$, it is possible to reconstruct the various components of the gap function.

## 1.7 SU free energy density

BCS essentially relies on the insurgence of an effective free fermionic field, described by the Bogoliubov bands, and the condensation of Cooper pairs. Superconducting state is favored by the lowering of free energy granted by the shifting of much of the spectral weight to a lower-energy maximum [**zhou1992density**]. "Condensation energy" is simply the energy difference between the superconducting state and the normal state, and its negativity indicates instability towards superconducting order. It's pretty straightforward to derive the ground state energy for the standard BCS state. When performing Wick's contractions,

$$\begin{aligned}\langle\hat{c}^\dagger_\alpha\hat{c}^\dagger_\beta\hat{c}_\gamma\hat{c}_\delta\rangle \sim\ & \langle\hat{c}^\dagger_\alpha\hat{c}^\dagger_\beta\rangle\hat{c}_\gamma\hat{c}_\delta + \hat{c}^\dagger_\alpha\hat{c}^\dagger_\beta\langle\hat{c}_\gamma\hat{c}_\delta\rangle - \langle\hat{c}^\dagger_\alpha\hat{c}^\dagger_\beta\rangle\langle\hat{c}_\gamma\hat{c}_\delta\rangle && \text{Cooper terms}\\ & - \langle\hat{c}^\dagger_\alpha\hat{c}_\gamma\rangle\hat{c}^\dagger_\beta\hat{c}_\delta - \hat{c}^\dagger_\alpha\hat{c}_\gamma\langle\hat{c}^\dagger_\beta\hat{c}_\delta\rangle + \langle\hat{c}^\dagger_\alpha\hat{c}_\gamma\rangle\langle\hat{c}^\dagger_\beta\hat{c}_\delta\rangle && \text{Fock terms}\\ & + \langle\hat{c}^\dagger_\alpha\hat{c}_\delta\rangle\hat{c}^\dagger_\beta\hat{c}_\gamma + \hat{c}^\dagger_\alpha\hat{c}_\delta\langle\hat{c}^\dagger_\beta\hat{c}_\gamma\rangle - \langle\hat{c}^\dagger_\alpha\hat{c}_\delta\rangle\langle\hat{c}^\dagger_\beta\hat{c}_\gamma\rangle && \text{Hartree terms}\end{aligned}$$

Form each of the three rows above, the last term is the "leftover" part which shifts energy. Thus it can be ignored in derivation, but becomes important when considering the energy in the end. The EHM hamiltonian can be written as:

$$\begin{aligned}\hat{H} = \sum_{\mathbf{k}}\xi_{\mathbf{k}}\left[\hat{c}^\dagger_{\mathbf{k}\uparrow}\hat{c}_{\mathbf{k}\uparrow} + \mathbb{I} - \hat{c}_{-\mathbf{k}\downarrow}\hat{c}^\dagger_{-\mathbf{k}\downarrow}\right] - \sum_{\mathbf{k},\mathbf{k}'}\left[V^{(s)} + V_{\mathbf{k}\mathbf{k}'}\right]\hat{\phi}^\dagger_{\mathbf{k}}\hat{\phi}_{\mathbf{k}'}\\ - \frac{2V}{L_xL_y}\sum_{\mathbf{k},\mathbf{k}'}\sum_{\sigma}\left[\cos(\delta k_x) + \cos(\delta k_y)\right]\hat{n}_{\mathbf{k}\sigma}\hat{n}_{\mathbf{k}'\sigma}\end{aligned} \tag{1.26}$$

as in Eq. (1.24). Following the scheme above, the "leftovers" of the MFT hamiltonian are just

$$\sum_{\mathbf{k},\mathbf{k}'}\Delta_{\mathbf{k}}\langle\hat{\phi}_{\mathbf{k}'}\rangle - \frac{2V}{L_xL_y}\sum_{\mathbf{k},\mathbf{k}'}\sum_{\sigma}\left[\cos(\delta k_x) + \cos(\delta k_y)\right]g_{\mathbf{k}}g_{\mathbf{k}'}$$

Further developing this result and applying Eqns. (**??**), (**??**), these last terms can be written as:

$$\sum_{\mathbf{k}}\frac{|\Delta_{\mathbf{k}}|^2}{2\tilde{E}_{\mathbf{k}}}\tanh\left(\frac{\beta\tilde{E}_{\mathbf{k}}}{2}\right) - 2VL_xL_y\sum_{\gamma}\left|g^{(\gamma)}\right|^2 \qquad \gamma\in\{s^*, d_{x^2-y^2}\} \tag{1.27}$$

as for Eq. (**??**). Notice the presence of a volume factor in the $\mathcal{O}\left(g^2\right)$ correction.

Now, we retrieve the expression for the MFT free energy density [**hutchinson2021mixed**] simply by adding four contributions:

1. Bogoliubov free quasiparticles thermal ground state energy density,
$$\frac{1}{L_xL_y}\sum_{\mathbf{k}}\tilde{E}_{\mathbf{k}}\left[f(\tilde{E}_{\mathbf{k}};\beta,0)-f(-\tilde{E}_{\mathbf{k}};\beta,0)\right]=\frac{1}{L_xL_y}\sum_{\mathbf{k}}\tilde{E}_{\mathbf{k}}\left[f(\tilde{E}_{\mathbf{k}};\beta,0)-\Big(1-f(\tilde{E}_{\mathbf{k}};\beta,0)\Big)\right]$$
$$=\frac{2}{L_xL_y}\sum_{\mathbf{k}}\tilde{E}_{\mathbf{k}}f(\tilde{E}_{\mathbf{k}};\beta,0)-\frac{1}{L_xL_y}\sum_{\mathbf{k}}\tilde{E}_{\mathbf{k}}$$
having used the simple property of the Fermi-Dirac distribution $f(-x;\beta,0)=1-f(-x;\beta,0)$.
2. The term originated by the identity produced by fermionic anticommutation rules in Eq. (1.26),
$$\frac{1}{L_xL_y}\sum_{\mathbf{k}}\tilde{\xi}_{\mathbf{k}}$$
averaged on the lattice.
3. The "leftovers" of Eq. (1.27), divided by $L_xL_y$.
4. The entropic free energy
$$-Ts$$
with $T=1/k_B\beta$ and $s$ the free Fermi gas entropy density for the Bogoliubov quasiparticles,
$$s=-\frac{2k_B}{L_xL_y}\sum_{\mathbf{k}}\left[\Big(1-f(\tilde{E}_{\mathbf{k}};\beta,0)\Big)\ln\Big(1-f(\tilde{E}_{\mathbf{k}};\beta,0)\Big)+f(\tilde{E}_{\mathbf{k}};\beta,0)\ln f(\tilde{E}_{\mathbf{k}};\beta,0)\right]$$
We use the identity:
$$\ln\left(1-\frac{1}{e^x+1}\right)=x+\ln\left(\frac{1}{e^x+1}\right)$$
and reduce the above expression to:
$$s=-\frac{2k_B}{L_xL_y}\sum_{\mathbf{k}}\Big[\ln\Big(1-f(\tilde{E}_{\mathbf{k}};\beta,0)\Big)+f(\tilde{E}_{\mathbf{k}};\beta,0)\ln f(\tilde{E}_{\mathbf{k}};\beta,0)$$
$$-\beta\tilde{E}_{\mathbf{k}}f(\tilde{E}_{\mathbf{k}};\beta,0)-f(\tilde{E}_{\mathbf{k}};\beta,0)\ln f(\tilde{E}_{\mathbf{k}};\beta,0)\Big]$$
Second and fourth terms cancel out, while the third compensates the first one of contribution 1.

All together, the MFT averaged free energy looks like this:
$$f_{\mathrm{MFT}}[\Delta_{\mathbf{k}},g_{\mathbf{k}}]=\frac{1}{L_xL_y}\sum_{\mathbf{k}}\left[\tilde{\xi}_{\mathbf{k}}-\tilde{E}_{\mathbf{k}}+\frac{|\Delta_{\mathbf{k}}|^2}{2\tilde{E}_{\mathbf{k}}}\tanh\left(\frac{\beta\tilde{E}_{\mathbf{k}}}{2}\right)\right]$$
$$-2V\sum_{\gamma\in\{s^*,d\}}\left|g^{(\gamma)}\right|^2+\frac{2}{\beta L_xL_y}\sum_{\mathbf{k}}\ln\Big(1-f(\tilde{E}_{\mathbf{k}};\beta,0)\Big)\tag{1.28}$$

The two main differences with the commonly known expressions for $f_{\mathrm{MFT}}$ [**hutchinson2021mixed**, **senarath2025mixed**] are the presence of bands renormalization , the absence of $-\mu n$ [Why?], and the $\mathcal{O}\left(g^2\right)$ correction. The latter, not only is negative for attractive $V$, a feature that can lead to superconducting phase energy lowering and stabilisation, but also it depends explicitly on the spectral amplitude of the $s^*$ and $d$-wave components of $g_{\mathbf{k}}$. Since as seen the $d$-wave part of $g_{\mathbf{k}}$ arises for mixed symmetries, this details seems to indicate an energetic preference for multi-symmetry phases.

## 1.8 Numerical simulations

[To be continued...]

### 1.8.1 HF iterative algorithm sketch

[To be continued...]